

Experiment 18 — Perceptron-Based IRIS Classification

AIM

To implement IRIS flower classification using the Perceptron algorithm in Google Colab.

SOFTWARE USED

Google Colab

PROCEDURE

1. Open Google Colab and create a new notebook.
2. Import the Iris dataset from Scikit-learn.
3. Separate the input features and target classes.
4. Split the dataset into training and testing data.
5. Create a Perceptron classifier.
6. Train the Perceptron model using the training data.
7. Predict the Iris classes using the testing data.
8. Calculate the classification accuracy.
9. Display the actual and predicted classes.

CODE

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.linear_model import Perceptron
from sklearn.metrics import accuracy_score

# Load Iris dataset
iris = load_iris()

# Input features
X = iris.data

# Target classes
```

```
y = iris.target
# Split dataset
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)
# Create Perceptron model
model = Perceptron(
    max_iter=1000,
    random_state=42
)
# Train model
model.fit(X_train, y_train)
# Predict test data
y_pred = model.predict(X_test)
]
# Calculate accuracy
accuracy = accuracy_score(y_test, y_pred)
print("Perceptron Classification")
print("-----")
print("Accuracy:", accuracy)
# Display actual and predicted values
print("\nActual Classes:")
print(y_test)
print("\nPredicted Classes:")
print(y_pred)
```

OUTPUT

Perceptron Classification

Accuracy: 0.6333333333333333

Actual Classes:

[1 0 2 1 1 0 1 2 1 1 2 0 0 0 0 1 2 1 1 2 0 2 0 2 2 2 2 0 0]

Predicted Classes:

[1 0 1 1 1 0 1 1 1 1 1 0 0 0 0 1 1 1 1 1 0 1 0 1 1 1 1 1 0 0]

RESULT

Thus, the IRIS flower classification was successfully implemented using the Perceptron algorithm in Google Colab, and the classification accuracy was evaluated.